

Module 01: Systems — Probability and Generative Models

Course: Active Inference 101 | **Unit:** Mathematical Frameworks | **Audience:**
First-semester undergraduates

Learning Objectives

1. Review **probability basics**: random variables, probability distributions, conditional probability, Bayes' theorem.
2. Define a **generative model** mathematically: a joint distribution $P(\text{observations, hidden states})$.
3. Explain why generative models are the mathematical formalization of "systems" in Active Inference.

Introduction

The Cognitive Science and Computational Neuroscience units introduced Active Inference conceptually and neurally. Now we develop the *mathematics*. Don't worry — we'll build up from basics. This module starts with probability theory and generative models.

Key Concepts

1. Random Variables and Probability Distributions

A **random variable** is a quantity whose value is uncertain. We describe uncertainty with **probability distributions**:

- **Discrete**: $P(X = x)$ for each possible value (e.g., rolling a die: $P(X = 3) = 1/6$)
- **Continuous**: $p(x)$ — a probability density function (e.g., human height follows a bell curve)

Key properties:

- All probabilities are between 0 and 1
- Total probability sums/integrates to 1

2. Joint, Marginal, and Conditional Probability

For two random variables X and Y :

- **Joint:** $P(X, Y)$ — probability of X *and* Y occurring together
- **Marginal:** $P(X) = \sum_Y P(X, Y)$ — probability of X regardless of Y
- **Conditional:** $P(X | Y)$ — probability of X *given* that Y is known

Example: X = "it's raining," Y = "you see dark clouds"

- $P(\text{rain, clouds})$ = probability of both
- $P(\text{rain})$ = total probability of rain
- $P(\text{rain} | \text{clouds})$ = probability of rain given you see clouds (higher than $P(\text{rain})$ alone!)

3. Bayes' Theorem — The Update Rule

Bayes' theorem is the mathematical engine of Active Inference:

$$P(s | o) = P(o | s) \times P(s) / P(o)$$

Where:

- $P(s | o)$: Posterior — what we believe about hidden state s after observing o
- $P(o | s)$: Likelihood — how probable observation o is given state s
- $P(s)$: Prior — what we believed before observing anything
- $P(o)$: Evidence — how probable the observation is overall (normalizing constant)

Everyday example: You hear a scratching sound (o). Was it your cat (s_1) or a burglar (s_2)?

- Prior: $P(\text{cat}) = 0.95$, $P(\text{burglar}) = 0.05$ (cats scratch more often)
- Likelihood: $P(\text{scratch} | \text{cat}) = 0.8$, $P(\text{scratch} | \text{burglar}) = 0.3$
- Posterior: $P(\text{cat} | \text{scratch}) \approx 0.98$ — even more certain it's the cat

4. Generative Models — The Core Formalism

A **generative model** is a joint probability distribution that specifies how hidden states generate observations:

$$P(o, s) = P(o | s) \times P(s)$$

This says: "The world (state s) generates observations (o) according to likelihood $P(o | s)$, and states have prior probability $P(s)$."

The generative model is the mathematical version of the brain's "model of the world" from the Cognitive Science unit. It contains:

- What states exist (hidden states s)
- How states generate observations (likelihood)
- What the agent expects in advance (prior)

5. From Static to Dynamic: Adding Time

Real systems change over time. We extend the generative model:

$$P(o_{1:T}, s_{1:T}) = P(s_1) \times \prod_t P(s_t | s_{1:t-1}) \times \prod_t P(o_t | s_t)$$

This is a **Hidden Markov Model (HMM)** — the simplest dynamic generative model:

- $P(s_1)$: Initial state distribution
- $P(s_t | s_{1:t-1})$: Transition dynamics (how states evolve)
- $P(o_t | s_t)$: Observation model (how states generate data)

Summary

Probability theory provides the language for Active Inference. Generative models — joint distributions over observations and hidden states — formalize the brain's model of the world. Bayes' theorem explains how agents update beliefs when new data arrives. Hidden Markov Models extend this to dynamic systems.

Further Reading

- Bishop, C. M. (2006). *Pattern Recognition and Machine Learning*. Springer. (Chapter 2)
- MacKay, D. J. C. (2003). *Information Theory, Inference, and Learning Algorithms*. Cambridge University Press.
- Parr, T., Pezzulo, G., & Friston, K. J. (2022). *Active Inference*. MIT Press. (Chapter 2)